Accident Pattern and Risk Intelligence System in India (2018-2020)

1. Introduction

Road accidents are a major public safety issue in India, leading to a high number of deaths and injuries every year. Understanding accident patterns helps in identifying risk factors and improving road safety measures. This project analyzes accident data from 2018, 2019, and 2020 to study trends, severity, and high-risk categories.

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2. Objective

- To analyze accident data from 2018–2020
- To identify trends in victims and offenders
- To calculate accident severity rate
- To find high-risk states and transport types
- To provide suggestions for improving road safety

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3. Dataset Description

- Sample size: 500 rows (sample dataset)
- Years covered: 2018, 2019, 2020
- Data includes:
    - Number of victims
    - Number of offenders
    - Transport type
    - State-wise accident data

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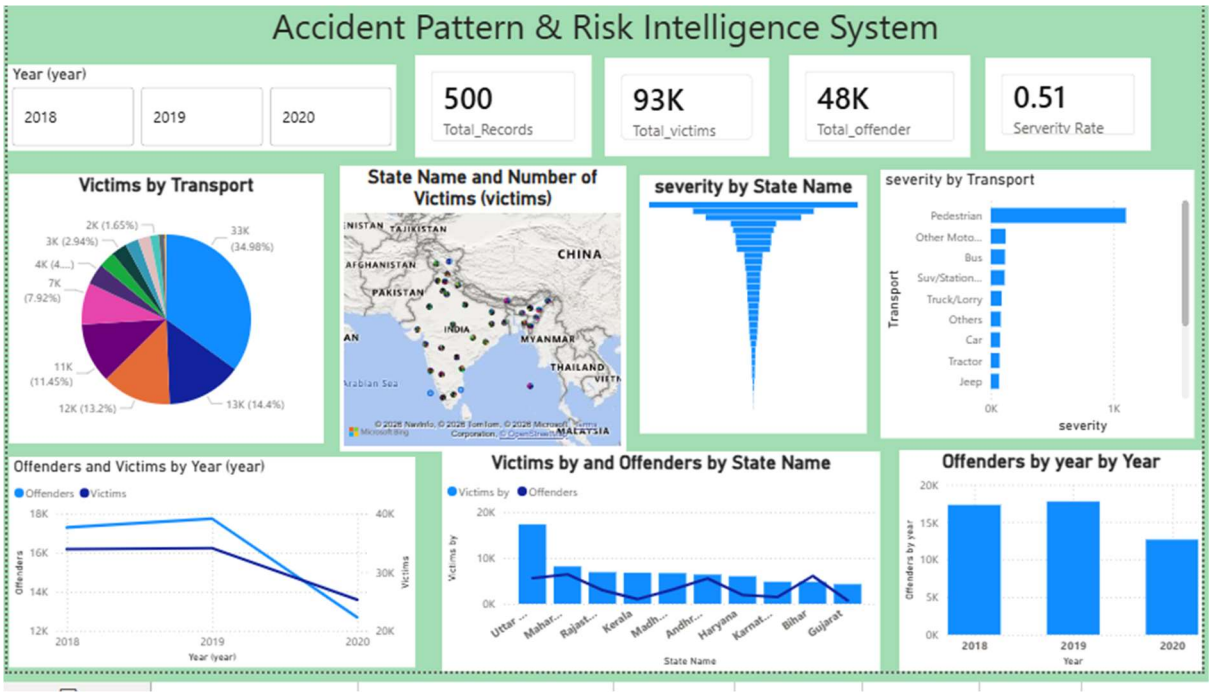
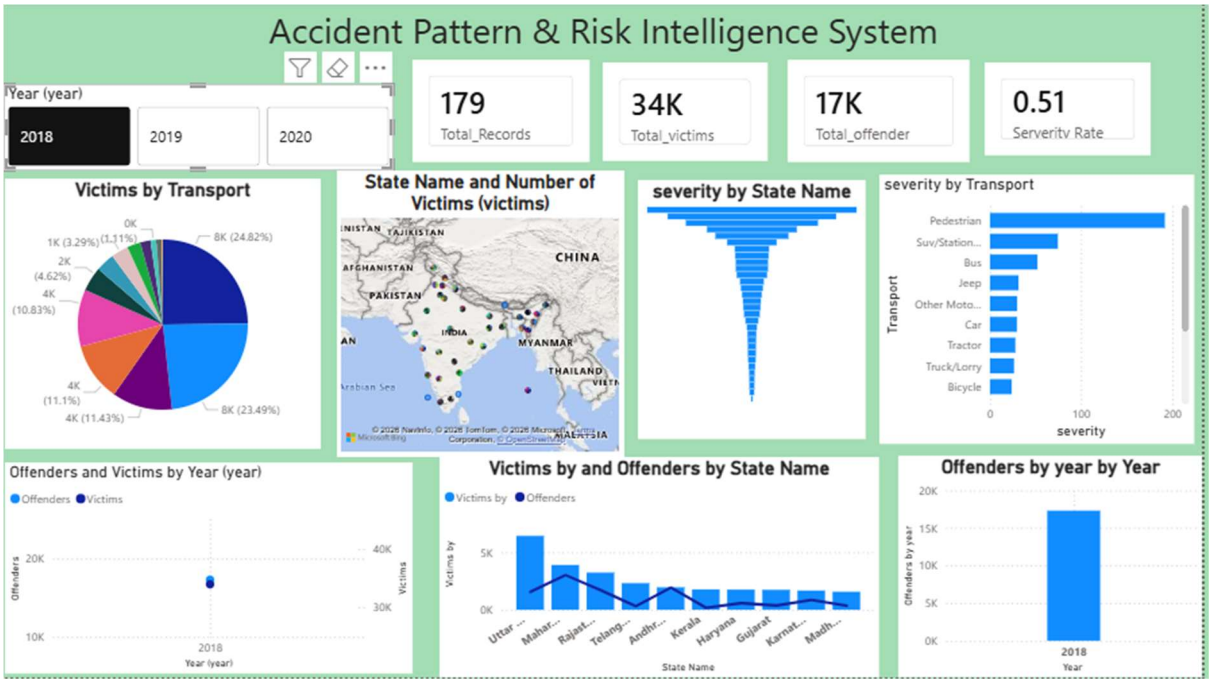
4. Methodology

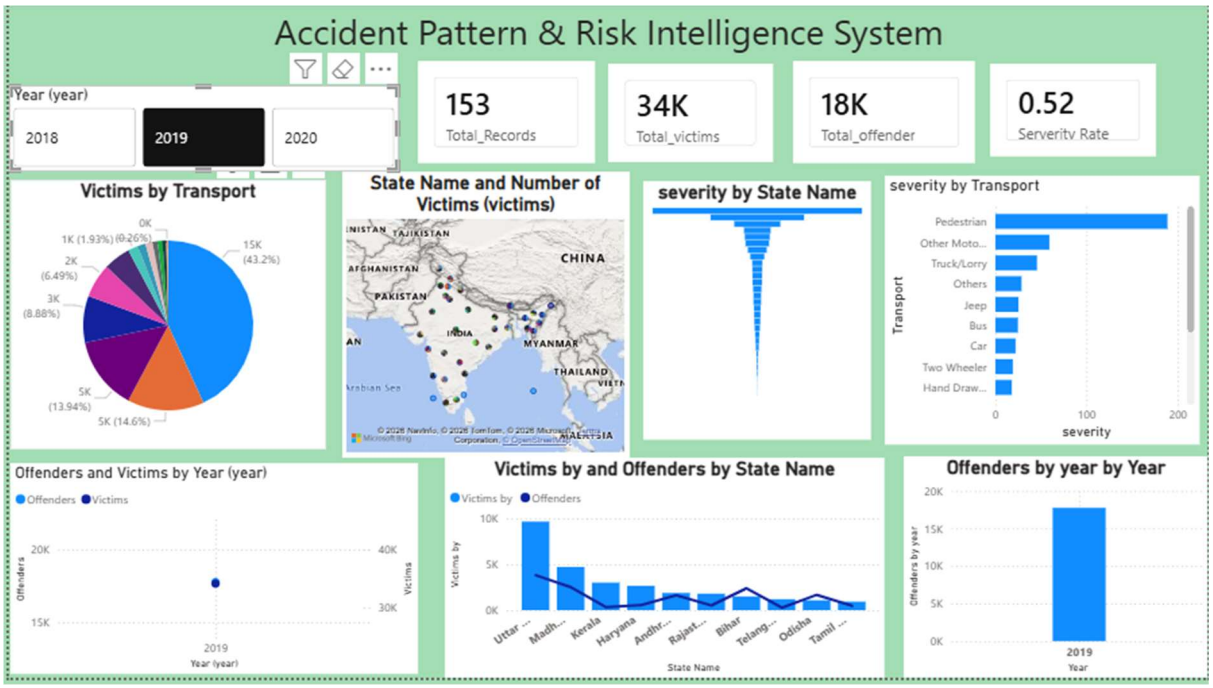
- Data cleaning and preprocessing
- Calculation of severity rate
- Year-wise comparison
- Transport-wise analysis
- State-wise analysis

Severity Rate Formula:

Severity Rate = Number of Victims / Number of Offenders







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5. Analysis and Findings

5.1 Descriptive Analysis

- Total victims:
  - 2018: 34,000
  - 2019: 24,000
  - 2020: 24,000

- Total offenders:
  - 2018: 17,000
  - 2019: 18,000
  - 2020: 13,000

Observation:
There is a decrease in victims after 2018. The number remains stable in 2019 and 2020.

5.2 Severity Analysis

- Severity Rate:
  - 2018: 0.51
  - 2019: 0.52
  - 2020: 0.50

Observation:
The severity rate is consistently around 0.5, indicating that accident impact remains high.

5.3 Transport-wise Analysis

- 2018:
  - Truck: 24.82%
  - Two-wheeler: 23.49%
- 2019:
  - Major category: 43.2%
- 2020:
  - Two-wheeler: 39.1%
- Highest severity transport type:
  - Pedestrians (all years)

Observation:
Two-wheelers and pedestrians are the most vulnerable road users.

5.4 State-wise Analysis

- Highest severity states:

- 2018: Gujarat
- 2019: Kerala
- 2020: Karnataka

Observation:
Different states show high severity in different years, indicating regional variation in accident risk.

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6. Conclusion

- Accident severity remains high across all years
- Pedestrians are the most affected group
- Two-wheelers contribute significantly to accidents
- High-risk states vary each year

This analysis helps in understanding accident trends and identifying critical risk areas.

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7. Recommendations

- Improve pedestrian safety (footpaths, crossings)
- Enforce helmet usage for two-wheeler riders
- Implement strict traffic rules in high-risk states
- Use data-driven systems for accident prediction

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8. Future Scope

- Use full dataset instead of sample data
- Apply machine learning models for prediction
- Include more factors like weather and road conditions
- Develop a real-time risk intelligence system

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9. References

- Government road accident reports (India)
- Public datasets on road safety